

ColliderAgent revision: audit of open placeholders

measurements of 2026-09-16/17 and proposed text

2026-09-17

Legend

DONE = measured today, text below can be pasted. **PARTIAL** = partly measured; the rest is in progress. **RUNS** = needs the remaining benchmark runs (Opus 4.6 $\times$ 3 per benchmark; cross-model). **AUTHOR** = a decision or confirmation only the authors can make.

1 Table S2 (sm.pdf p.4) — software environment

Status: **DONE** for every package (all read inside the images); only the Claude Code version of the *original* runs is open (see note). All versions were read inside the images on zhustation on 2026-09-16 with a probe blueprint (**collider-env-probe**; job logs are cited per row). Two corrections to the current table: the analysis Python stack lives in the *agent runtime*, not in the collider image (the image has no **uproot**), and Delphes is 3.5.1.

Image / runtime	Package	Version
collider	MadGraph5_aMC@NLO	3.7.0
collider	Pythia8	8.316
collider	Delphes	3.5.1
collider	MadAnalysis5	1.11.0 (2025-04-23)
collider	LHAPDF	6.5.5
collider	HepMC	2.06.09
collider	ROOT	6.28/10
collider	Python	3.10.12; numpy 1.26.4, scipy 1.15.3, awkward 2.9.0, matplotlib 3.10.8
mma-het	Wolfram Engine	13.3.0
mma-het	FeynRules	2.3.49 (29 Sep 2021)
micromegas	micrOmegas / CalcHEP	6.3.0 / 3.9.2
agent runtime	Python analysis stack	Python 3.13.12; uproot 5.7.4, awkward 2.9.0, numpy 2.4.4, scipy 1.15.3
agent runtime	Claude Code	2.1.273 (re-runs of 2026-09-16); original runs: see note
agent runtime	LLM	Claude Opus 4.6 (claude-opus-4-6), still served on 2026-09-16

Proposed replacement rows (booktabs, same layout as the current table):

```
collider    & MadGraph5_aMC@NLO & 3.7.0 \\
            & Pythia8      & 8.316 \\
            & Delphes      & 3.5.1 \\
            & MadAnalysis5  & 1.11.0 \\
            & LHAPDF       & 6.5.5 \\
```

```

& HepMC                & 2.06.09 \\
& ROOT                  & 6.28/10 \\
& pyhf                   & 0.7.6 \\
mma-het    & Wolfram Engine   & 13.3.0 \\
            & FeynRules           & 2.3.49 \\
agent runtime & Claude Code      & 2.1.x (AUTHORS: version of the original runs from the usage logs
                                the 2026-09-16 re-runs used 2.1.273) \\
            & LLM                  & Claude Opus 4.6 \\
            & uproot, awkward, numpy, scipy & 5.7.4, 2.9.0, 2.4.4, 1.15.3 \\

```

Note on the Claude Code version: no session log from the April–May runs exists on this machine (the earliest local logs are from August 2026), so the original version has to come from the co-author who ran them (**AUTHOR**). The rows above are otherwise complete.

2 sm.pdf p.5, “LLM inference accounts for a minor fraction [AUTHORS: fill in numbers ...]”

Status: **PARTIAL** — measured for the ALP EFT benchmark with three current models; the Opus 4.6 numbers for all eight benchmarks are being produced now. The claim as written is *not* supported for the light benchmarks: attributing every gap before a model message to inference and every gap before a tool result to tool execution (main session plus stage subagents), the model share of wall-clock was 38% (Opus 5), 54% (Opus 4.8) and 51% (Sonnet 5); time inside `magnus run/status` calls was 60%, 45% and 47% (the two overlap when a stage polls a running job). Proposed wording, to be finalised with the Opus 4.6 numbers:

For the light benchmarks of Table S3 the wall-clock time splits roughly evenly between LLM inference and cluster jobs (model share XX–YY%); for the Dark-SMEFT campaigns, where each campaign submits more than 100 jobs, cluster time dominates.

3 Table S3 (sm.pdf p. 6) — resource usage; also paper.pdf p. 5 and p. 7, reply.pdf p. 4 and p. 6

Status: **PARTIAL** — one run of every benchmark with Claude Opus 4.6 (effort **xhigh**, v2 skills, cold memory, launched 2026-09-16 22:00 UTC); seven of nine prompts completed, two are TBD (footnotes a, b) and are handed over. Tokens = main session + all sub-agents (uncached + cache-write + cache-read); files = distinct paths written or edited; Magnus jobs = `magnus run` submissions; USD from the CLI’s `total_cost_usd`.

Benchmark	Wall [h]	Sub-agents	Magnus jobs	Files	Tokens in [M]	Tokens out [k]	USD	Re
Scalar LQ, m_{ej} [Fig. 2(a)]	6.41	2	42	28	214.0	220	–	TI
ALP EFT, $\not{E}_T$ [Fig. 2(b)]	0.86	3	6	17	36.9	56	22	su
U(1)’ scan [Fig. 2(c)]	1.05	3	17	25	5.1	16	9	su
U1 LQ mono- τ [Fig. 2(d)]	4.00	2	29	19	181.4	219	106	TI
Heavy N [Fig. S2(a,b)]	0.40	3	7	18	3.2	13	4	su
General Z' [Fig. S2(c)]	0.66	3	7	16	19.4	38	13	su
KK graviton [Fig. S2(d)]	1.35	2	16	14	4.2	51	6	su
U1 LQ at MuC, η [Fig. S2(e)]	0.55	0	14	2	1.3	5	6	su
U1 LQ at MuC, reach [Fig. S2(f)]	1.41	3	25	23	27.3	52	21	su

* end-to-end reproduction with a quantitative deviation recorded as a footnote (see §Footnotes).

^a **TBD (handed over)**. Run stopped by the operator after 6.4h without a figure. Model, validation and the 100 000-event LUXlep parton sample (lepton→photon LHE workaround) were done; two Pythia8+Delphes launches of the full sample finished on the cluster but failed at upload with “No space left on device” (the Delphes ROOT file alone is 27 GB against the job’s 10 GB packing space); a third launch that deletes HepMC/ROOT after the run (MG5 ! shell lines after the final **done**, keeping the LHCO) was running when stopped. The v2 skills now document the limit and the cleanup (commit 61800a9); a rerun is expected to take 1.5–2h.

^b **TBD (handed over)**. Attempt 1 generated all 18 Delphes runs (4.0h, 106 USD; four 9-run launches were lost to the same disk limit before the agent split them), then the orchestrator ended its turn with “waiting for event file downloads” and the headless session terminated without the analysis stage (failure mode (iv) in the paper’s classification). A continuation session in the same workspace produced the figure in 1.7 h / 29.5 USD with the reference’s shape but a 20–35 % weaker exclusion. The orchestrator skill now forbids ending a turn to wait (commit e7c5b61).

Model-latency share of wall-clock (transcript timestamps): 0.33–0.80 across the nine runs, above 0.6 in seven of them; the sentence “LLM inference accounts for a minor fraction” (sm.pdf p.5) should be replaced by, e.g.:

For the figure benchmarks of Table S3 the wall-clock time is shared between LLM inference and cluster jobs (model share 30–80 %, median 0.68); cluster time dominates only for the Delphes-level benchmarks and the Dark-SMEFT campaigns.

4 Tables S4 and S5 — state and plan (handed over)

Current counts (one attempt per cell unless noted; Opus 4.6 at **xhigh**, Opus 4.8 / Opus 5 / Sonnet 5 at default effort):

Benchmark	Opus 4.6	Opus 4.8	Opus 5	Sonnet 5
Scalar LQ	TBD ^a	–	–	–
ALP EFT	1/1	1/1	1/1	1/1
U(1)’ scan	1/1*	–	1/1*	–
U1 LQ mono- τ	TBD ^b	–	–	–
Heavy N	1/1	–	–	–
General Z’	1/1	–	–	–
KK graviton	1/1*	–	–	–
U1 LQ at MuC (2 figures)	2/2*	–	–	–

Success criterion used (agreed 2026-09-16): the run completes end to end and the figure agrees with the requested part of the reference qualitatively; quantitative deviations go to a footnote (*). Failure modes follow the paper’s four classes; every verdict is in `bench_runs/<label>/verdict.yaml` with the assistive judge’s reasoning and a human note.

Plan to complete S4/S5:

1. **S5 (Opus 4.6 ×3)**. Two more attempts per benchmark: `scripts/bench/run_benchmark.sh <arxiv> <fig> claude-opus-4-6 --effort high` (**high** rather than **xhigh**: the two Delphes benchmarks cost 106 USD and 200 USD at **xhigh**, the others 4–22 USD). Launch at most 4 in parallel (cluster and 10 GB job storage). Then `judge.sh` with the reference PNGs listed in the handover, confirm each verdict, `aggregate.py`.
2. **S4 column B = GPT-5.3-Codex via Codex** on branch `codex-com` (skills-only mode; same day of release as Opus 4.6, 2026-02-05). Same prompts, one clean directory per attempt, same judge; record wall-clock, jobs and tokens from the Codex session logs.

3. **S4 column C = Claude Sonnet 5** (or Opus 5) with the harness, one attempt per benchmark first.
4. Fill the two TBD cells: rerun Scalar LQ and mono- τ with the current skills (both root causes are fixed in the skills; expected 1.5–2 h each).

5 Other markers

Where	Marker	Status	Proposed action
paper.pdf p. 5	“skills-only mode executes the hello-world workflow of Sec. S-III [confirm that this has been tested]” (Codex/Cursor)	AUTHOR	The Codex-compatible layout exists (branch <code>codex-com: .agents/skills, .codex/agents, AGENTS.md</code>). Run the hello-world prompt with Codex on that branch once and keep the sentence; otherwise weaken to “is designed to”.
paper.pdf p. 7, Table I caption	“tokens consumed and wall-clock time [fill in]”	RUNS	Same numbers as Table S3 (Opus 4.6 column); cite Table S3 instead of repeating them.
paper.pdf p. 9–10	“[confirm the extent of human intervention between campaigns]”	AUTHOR	Dark-SMEFT campaigns were not rerun; statement must come from the campaign logs.
paper.pdf p. 10; reply.pdf p. 6	“several weeks of work for an experienced phenomenologist [adjust estimate]”	AUTHOR	Judgement call; unchanged by measurements.
reply.pdf p. 1	checklist (1)–(4): fill S3/S4/S5, confirm human intervention, set <code>\markupfalse</code> , remove red markers	PARTIAL	(1) in progress, (2) author, (3)–(4) mechanical once (1)–(2) are done.
reply.pdf p. 4	“the LLM token consumption [fill in from the usage logs]”	RUNS	From Table S3; note the main+sub-agent definition.
reply.pdf p. 4	“The wall-clock time of a benchmark (1–6 hours) ...”	DONE	Consistent with today’s runs (0.9–1.4 h for the lightest benchmark); keep.
sm.pdf p. 4	“archived release ... DOI and per-run provenance packaged as Workflow Run RO-Crates [confirm or remove]”	AUTHOR	No RO-Crate export exists in the repository today; either remove or keep as a plan.
sm.pdf p. 5	Table S4 placeholder text “e.g. Claude Opus 4.6 vs. Claude Sonnet, GPT-5.x via Codex, Gemini via ADK”	PARTIAL	Replace with the three columns chosen above; drop Gemini/ADK unless the python-agent path is run.
sm.pdf p. 6	Table S3 Dark-SMEFT rows (sub-agent calls, tokens)	AUTHOR	Not reproducible here; from the campaign logs.

6 Provenance of today’s numbers

Branch `feat/skill-evolution-v2` of the ColliderAgent repository: `docs/paper-tbd-audit-2026-09-16.md` (measurements and observations), `bench_runs/<label>/metrics.json` and `verdict.yaml` per run, `scripts/bench/`

for the tables, probe blueprint **collider-env-probe** on zhustation for the image versions. Skills used for the re-runs are the v2 skills (context-lean rewrite of 2026-09-16), not the April versions; the Opus 4.6 re-runs therefore measure the current system.